Frederick Kwame Minta

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Research Interests

- Robot perception under hardware-induced degradation from power, thermal and computing limits
- Perception-aware motion planning that accounts for the robot's own physical state
- Active perception and onboard autonomy for resource-constrained robots
- Multi-sensor fusion and state estimation robust to sensor dropout

Education

University of Mines and Technology (UMaT) **Jan. 2020 – Nov. 2024**
B.Sc. Computer Science and Engineering *Tarkwa, Ghana*
Selected coursework: Linear Algebra; Signals and Systems; Digital Signal and Image Processing; Simulation and Modelling; Embedded System Design; Control Systems; Robotics; Artificial Intelligence; Data Structures and Algorithms.

Research Experience

Research Assistant, Robotics Research Platform **Nov. 2025 – Present**
Ashesi University | Supervisor: Dr. Ayorkor Korsah *Berekuso, Ghana*
- Lead hardware development for the Ashesi Robot Application Platform (ARAP), a 1.55 m, 42 kg open-source differential-drive robot: custom ATmega2560-compatible controller board, power distribution, motor drivers, sensor integration and bolted chassis.
- Develop the ROS 2 and Gazebo navigation stack and the multi-sensor fusion pipeline for state estimation; validated on indoor delivery and tour-guiding tasks.
- Diagnosed a hardware-induced perception failure: motor inrush pulled the supply rail down about 1.5 V and dropped the LiDAR for roughly 3 s. After mitigation, 50 consecutive dual-motor starts produced no dropout.
- Produced the thermal and power budget for 28–35 °C ambient operation, placing vents and fans from a CFD study; a 30-minute run at 70% motor duty held every monitored component at least 20 °C below limit.
- Lead the open-source release under the CERN Open Hardware Licence, preparing design files, BOM and build documentation.

Research Assistant, Assistive Technology Lab **Oct. 2024 – Aug. 2025**
University of Ghana | Supervisor: Dr. Millicent Agangiba *Accra, Ghana*
- Led a team of four researchers developing assistive robotics for people with upper-limb amputations.
- Built a ROS 2 and OpenCV gesture-recognition interface for hands-free device control, reaching 94% classification accuracy in trials with upper-limb amputee participants.
- Implemented Deep Q-Networks in PyTorch for adaptive prosthetic control, reducing per-user training time by 60% relative to a non-adaptive baseline.
- Designed and ran human-in-the-loop studies with amputee participants, covering protocol, data collection and analysis.
- Integrated human motion and intent cues into the control loop, aligning assistance with user behaviour rather than requiring the user to adapt to a fixed controller.

IoT and Machine Learning Researcher **Sep. 2024 – Aug. 2025**
Farmlynco *Legon, Ghana*
- Built machine-learning pipelines for crop-yield prediction from multi-sensor field data, weather APIs and satellite-derived features, working with noisy, sparse and intermittently transmitted data.
- Developed decision-support tools for irrigation scheduling, producing 30% water savings in deployment.

Research Assistant, Hybrid Presplit Drilling and Blasting Project **Aug. 2023 – Jul. 2024**
University of Mines and Technology | Supervisor: Dr. Thompson Brantson *Tarkwa, Ghana*
- Designed PCB-based wireless detonation systems using ESP32 microcontrollers and mesh networking, reducing inter-hole delay error by 40% in field tests.
- Developed safety-critical firmware with fail-safe behaviour for loss of communication or power, and contributed to patent-related documentation.

Research Assistant, Agri-IoT Lab **Nov. 2023 – Jun. 2024**
University of Mines and Technology | Supervisor: Dr. Emmanuel Effah *Tarkwa, Ghana*
- Deployed convolutional models with TensorFlow Lite on Raspberry Pi 4 for real-time crop-disease detection under on-board compute and power constraints.
- Architected an AWS IoT Core and DynamoDB pipeline processing more than 10,000 sensor readings daily, using MQTT over unreliable rural links.

Publications and Presentations

Under Review

[R1] **Minta, F. K.**, Sitti, M., Agangiba, M., Essile, A. Assistive Robot for Children with Disability (Amputated Arms). *IEEE Transactions on Robotics*. Under review.

In Preparation

[P1] **Minta, F. K.**, Takyi, J., Aweenagua, F., Korsah, A. ARAP: A Modular Open-Hardware Mobile Robot Platform for Resource-Constrained Research Environments. In preparation for *HardwareX*.

Journal Articles

[J1] **Minta, F. K.**, Legese, F. G., Alhassan, A. (2025). Generative AI in Child–Robot Interaction: An Adaptive Framework with Safety and Developmental Alignment. *Advanced International Journal for Research*, 6(5).

[J2] Aidoo, B., **Minta, F. K.**, N-yo, A.-A., Tettey, D. A., Kasiimbura, O., Essile, A. (2025). A Scalable IoT-Cloud Architecture with Deep Q-Learning for Air Quality Monitoring and Alerting. *International Journal of Computer Applications*, 187(23).

[J3] Okai, G. E. Y., **Minta, F. K.**, Osman, A.-M., Essile, A. (2024). IoT-based Weather Monitoring System for Ghanaian Farmers. *International Journal of Computer Applications*, 186(42).

Conference Papers

[C1] Aidoo, B., **Minta, F. K.**, N-Yo, A.-A., Tettey, D. A. (2025). Reinforcement Learning-Driven IoT System for Air Quality Monitoring and SMS Alerting. *5th IEEE ICECCME*, Zanzibar, pp. 1–6. Oral presentation.

Presentations

[T1] Aidoo, B., **Minta, F. K.** (2025). Cocoa Pod Maturity Stages for Crop Monitoring. Poster, *Deep Learning Indaba*, Kigali.

Research Funding

EPICS in IEEE Grant, Jon C. Taenzer Memorial Fund (IEEE Foundation, 2025). Project Lead: assistive technology enabling blind students to participate in STEM learning.

Teaching and Leadership

Co-Founder and Chief Technology Officer **Jan. 2024 – Present**
Coasted Code — AI school for children *Accra, Ghana*

- Built Coasted Code into an education provider accredited by STEM.org and named a Best in STEM Distinction honoree; deliver a project-based curriculum in embedded programming, sensor integration and control across urban and rural Ghana.

Technology Trainer, DARE Project **Aug. 2025 – Feb. 2026**
Inclusive Tech Group, with the Mastercard Foundation and UNESCO *Accra, Ghana*

- Trained 100 women with disabilities to use technology in their businesses, adapting instruction to varied accessibility and digital-literacy needs.

Robotics Instructor and Club President **Jun. 2023 – Sep. 2024**
Aaenics Engineering Limited and UMaT Aaenics Robotics Club *Ghana*

- Designed and delivered a robotics curriculum in embedded programming, sensor integration and PID control for more than 300 students across five institutions, with outreach to over 200 school pupils.

Technical Expertise

Robotics	ROS 2, Gazebo, SLAM, motion planning, sensor fusion, state estimation, OpenCV, deep reinforcement learning, human–robot interaction
Programming	Python, C++, C, MATLAB, JavaScript, SQL
Embedded	ATmega2560, ESP32, Raspberry Pi, NVIDIA Jetson Orin NX, PCB design (KiCad), C/C++ firmware, MQTT
Hardware	Power-system design, DC/DC converter integration, motor drivers, CFD thermal analysis, CAD
Machine learning	PyTorch, TensorFlow / TensorFlow Lite, scikit-learn, edge deployment under compute and power constraints

Honours

Best in STEM Distinction Honoree (Coasted Code), STEM.org	Sep. 2026
Winner, BeOrchid African Hackathon	Jul. 2026
Best Robotics Student, UMaT Computer Science and Engineering Awards	Sep. 2024
First Runner-Up, National Disability Inclusive Hackathon	Dec. 2021

Affiliations: Black in AI (ELAI Leadership Program); Black in Robotics. **Service:** Reviewer, IndabaX Ghana, 2024.